

EXERCISE 3: STORED PROCEDURES

Name: Tejaswi

TABLE CREATION

Question:

Create the required tables for Savings Accounts, Employees, and Accounts.

```
[ SQL Worksheet ]*  ▶  ⋮  🔑  📄  ⋮  Aa  🗑️

1 CREATE TABLE SavingsAccounts (
2     AccountID NUMBER PRIMARY KEY,
3     CustomerName VARCHAR2(50),
4     Balance NUMBER
5 );
6
7 CREATE TABLE Employees (
8     EmployeeID NUMBER PRIMARY KEY,
9     EmployeeName VARCHAR2(50),
10    DepartmentID NUMBER,
11    Salary NUMBER
12 );
13
14 CREATE TABLE Accounts (
15     AccountID NUMBER PRIMARY KEY,
16     CustomerName VARCHAR2(50),
17     Balance NUMBER
18 );
19
```

INSERTION

```
1 INSERT INTO SavingsAccounts VALUES (301, 'Tejaswi', 12000);
2 INSERT INTO SavingsAccounts VALUES (302, 'Ananya', 18000);
3
4 INSERT INTO Employees VALUES (11, 'Keerthana', 10, 50000);
5 INSERT INTO Employees VALUES (12, 'Harika', 10, 60000);
6 INSERT INTO Employees VALUES (13, 'Siri', 20, 55000);
7
8 INSERT INTO Accounts VALUES (401, 'Tejaswi', 10000);
9 INSERT INTO Accounts VALUES (402, 'Ananya', 5000);
10
11 COMMIT;
```

SCENARIO 1

The bank needs to process monthly interest for all savings accounts.

Question:

Write a stored procedure ProcessMonthlyInterest that calculates and updates the balance of all savings accounts by applying an interest rate of 1% to the current balance.

Procedure:

```
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1  CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
2  AS
3  BEGIN
4      UPDATE SavingsAccounts
5          SET Balance = Balance + (Balance * 0.01);
6
7      COMMIT;
8  END;
9  /
10 BEGIN
11     ProcessMonthlyInterest;
12 END;
13 /
14 SELECT * FROM SavingsAccounts;
```

Output:

Query result

Script output

DBMS output

Explain Plan

SQL history

Download

Execution time: 0.006 seconds

	ACCOUNTID	CUSTOMERNAME	BALANCE
1	301	Tejaswi	12241.2
2	302	Ananya	18361.8

SCENARIO 2

The bank wants to implement a bonus scheme for employees based on their performance.

Question:

Write a stored procedure UpdateEmployeeBonus that updates the salary of employees in a given department by adding a bonus percentage passed as a parameter.

Procedure:

```
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1 CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(
2     p_DepartmentID IN NUMBER,
3     p_BonusPercent IN NUMBER
4 )
5 AS
6 BEGIN
7     UPDATE Employees
8     SET Salary = Salary + (Salary * p_BonusPercent / 100)
9     WHERE DepartmentID = p_DepartmentID;
10
11     COMMIT;
12 END;
13 /
14 BEGIN
15     UpdateEmployeeBonus(10,10);
16 END;
17 /
18 SELECT * FROM Employees;
```

Output:

Download [i] Execution time: 0.008 seconds					
	EMPLOYEEID	EMPLOYEENAME	DEPARTMENTID	SALARY	
1	11	Keerthana	10	55000	
2	12	Harika	10	66000	
3	13	Siri	20	55000	

SCENARIO 3

Customers should be able to transfer funds between their accounts.

Question:

Write a stored procedure TransferFunds that transfers a specified amount from one account to another, checking that the source account has sufficient balance before making the transfer.

Procedure:

```

1 CREATE OR REPLACE PROCEDURE TransferFunds(
2     p_FromAccount IN NUMBER,
3     p_ToAccount IN NUMBER,
4     p_Amount IN NUMBER
5 )
6 AS
7     v_Balance NUMBER;
8 BEGIN
9     SELECT Balance
10    INTO v_Balance
11   FROM Accounts
12  WHERE AccountID = p_FromAccount;
13
14     IF v_Balance >= p_Amount THEN
15
16         UPDATE Accounts
17        SET Balance = Balance - p_Amount
18       WHERE AccountID = p_FromAccount;
19
20         UPDATE Accounts
21        SET Balance = Balance + p_Amount
22       WHERE AccountID = p_ToAccount;
23
24         COMMIT;
25
26         DBMS_OUTPUT.PUT_LINE('Transfer Successful');
27
28     ELSE
29         DBMS_OUTPUT.PUT_LINE('Insufficient Balance');
30     END IF;
31 END;
32 /
33

```

Procedure Execution

```

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1 BEGIN
2     TransferFunds(401,402,2000);
3     END;
4     /
5     SELECT * FROM Accounts;

```

Output:

Query result	Script output	DBMS output	Explain Plan	SQL history
Download Execution time: 0.006 seconds				
	ACCOUNTID	CUSTOMERNAME	BALANCE	
1	401	Tejaswi	8000	
2	402	Ananya	7000	